**Element I: Testing, Data Collection, and Analysis Report**
**Test 1: (Durability + Sturdiness) The bookstand must be durable and withstand repeated use.**
**Priority Level:** High
**Purpose:** To test how the stand holds up to repeated use, including leaning, pressing, and adjusting.
**Independent Variable:** Force Applied (Simulated Student Pressure)
**Control:** Same Pressure Applied each time, same location and angle.
**Procedure Summary:**
- Pressed down on the book stand ten times in the center and edges
- Checked for visible damage, loosening, or movement after each trial.

**Raw Data Sample:**

| **Trial #** | **Damage Observed?** | **Functionality Affected?** |
|---|---|---|
| **1-10** | **No** | **No** |

**Analysis:** Data showed the stand remained completely intact after repeated pressure in all 10 trials. No joints loosened and no materials showed signs of cracking or bending. The design exceeds the expectations for durability and supports regular use.
**Implications**
- Ideal for long-term classroom or library use
- No reinforcements needed unless used in extreme environments

**Next Steps:**
- Consider minor cosmetic refinements
- Extend durability testing to long-term daily usage simulation.

**Test 2: (Stability) The bookstand must be adjustable/moveable without using stability.**
**Priority Level:** High
**Purpose:** To evaluate how well the book stand hold books without being knocked over or unstable.
**Independent Variable:** Book size and Placement
**Dependent Variable:** Stability of Book Stand.
**Control:** Flat Surface and same base position.
**Procedure Summary:**

- Place books of different sizes and thicknesses at various positions on the stand (Centered and off centred)
- Observe whether the stand tipped, slid, or wobbled during 5 trials per book.

**Raw Data Sample:**
**Small Paperback:** Fully stable
**Medium Novel (centered):** Stable
**Medium Novel (off centered):** Slight tilt.
**Large Textbook (centered):** Stable
**Large Textbook (off centered):** Wobble
**Analysis:**The prototype was generally stable across most tests, but stability decreased when heavy books were placed off center. This shows that the stand meets basic stability requirements but may need refinement for unbalanced loads.
**Next Steps:** Widen the base or add non-slip feet to prevent wobbling with uneven weight distribution.